Global Quantum + AI Challenge 2026

Enterprise Challenge Statement

1. Challenge Title

Quantum-Enhanced Vision-Language-Action Models in Autonomous Driving and Robotics Applications

Develop quantum-enhanced or quantum-inspired methods that boost training efficiency, model compactness, model compression, or safe task execution of Vision-Language-Action models for real-time autonomous-driving or low-power robotic applications.

2. Executive Summary

The real-world problem

Vision-Language-Action Models (VLAMs) are rapidly becoming the backbone of autonomous systems — processing multimodal sensor data to produce planning and control outputs. Models such as Alpamayo-R1 [3] and π_0 [7] demonstrate the category's potential, but their 7B–10B parameter scale makes them prohibitively expensive to train and nearly impossible to deploy on embedded vehicle hardware or robot controllers without significant accuracy loss.

Why it matters

Automotive OEMs, Tier-1 suppliers, logistics operators, and robotics manufacturers need VLAM-grade perception and reasoning to run on in-vehicle compute units and robot controllers — not on cloud GPU clusters. Every reduction in model size and training cost directly lowers deployment barriers and shortens development cycles for safety-critical systems.

Why classical methods fall short

Standard compression (quantization, pruning, distillation) can reduce model size, but the compound constraint is the problem: safety-critical deployments require accuracy guarantees on long-tail sensor distributions — not just general benchmarks — while simultaneously satisfying real-time latency, power, and formal certification requirements. No single classical technique addresses model footprint, training cost, and verifiable safety guarantees together.

Where quantum and quantum-inspired methods may help

Quantum-Inspired Tensor Networks have shown strong compression potential while preserving accuracy [1]. Quantum algorithms offer improved scaling for high-dimensional linear algebra central to VLAM training [2]. Quantum RL circuits may improve alignment sample efficiency. Together, these represent a differentiated research direction with documented early evidence of advantage.

What participants must deliver

A reproducible quantum-enhanced or quantum-inspired component that demonstrates VLAM training efficiency, model compression, RL alignment sample efficiency, or formal control safety. Required deliverables: working prototype, quantitative results against the track-specific baseline, and a 4–8 page technical report.

3. Industrial and Operational Context

In Autonomous Driving, a VLAM fuses camera, LiDAR, and radar inputs with a goal state to produce a trajectory or control command. The hard real-time constraint is the processing time of 100ms per inference cycle. Safety compliance is governed by ISO 26262. Vehicles must handle long-tail scenarios like adverse weather, construction zones, unpredictable pedestrians, where model compactness and formal safety guarantees are certification requirements, not optional features.

In the robotics domain, a VLAM interprets visual scenes and natural language task commands to control a manipulator's arm or mobile base. Deployment targets are embedded robot controllers (Jetson-class platforms), and collaborative robots (cobot) compute units with strict memory and power envelopes. Safety compliance is governed by IEC 62061 and ISO 10218. Models must generalize across diverse objects, surfaces, and instruction phrasings, placing a high premium on efficiency without sacrificing task success rate.

The following key stakeholders indicate the scale of the problem:

- ML engineers building and fine-tuning VLAM architectures for platform-specific deployment
- System architects managing hardware budgets, power envelopes, and real-time constraints
- Safety engineers requiring formal or verifiable performance guarantees for regulatory certification

- Fleet operators and factory managers who depend on consistent low-latency model behavior at scale
- Regulators (NHTSA, EU AI Act, ISO TC22/TC299) overseeing certification of AI-driven safety-critical systems

4. Challenge Objective

Design a framework, implement a Proof of Concept, or improve a VLAM to verifiably increase training efficiency or compress model weights beyond current benchmarks.

4.1 Primary Objective

Demonstrate a clear, well-justified quantum-enhanced or quantum-inspired computational advantage over an accepted classical baseline. The advantage may take the form of:

- Reduced parameter count at equivalent or near-equivalent task accuracy on open-sourced benchmarks
- Faster training convergence (fewer GPU-hours or optimization steps) versus the accepted classical baseline
- Improved RL alignment sample efficiency — fewer rollouts to reach equivalent policy rewards
- Formal Lyapunov-based stability to guarantee mathematical safety assurance under perturbation

4.2 Secondary Objectives (optional)

- Energy efficiency: Report kWh or CO_2e consumed versus the classical baseline
- Cross-track generalizability: Show the method transfers from one track to the other
- Open-source release: Publish code and checkpoints under Apache 2.0 or MIT
- Hardware pathway: Simulation-based analysis of near-term quantum hardware feasibility (qubit count, circuit depth, noise sensitivity)

5. Problem Definition

5.1 Common VLAM Bottlenecks

The Challenge involves four core bottlenecks. Participants may target one or more; addressing multiple with an integrated approach is eligible for bonus recognition.

Bottleneck	Why It Is Hard to Solve Classically	How Quantum / QI Methods May Help
Model Footprint (7B–10B params)	VLAMs at this scale cannot run on embedded vehicle SoCs (e.g., NVIDIA Orin) or robot controllers. Memory overhead is prohibitive, and standard compression (quantization, pruning, distillation) degrades accuracy unacceptably beyond certain compression.	Quantum-Inspired Tensor Networks — specifically Matrix Product States (MPS) and Tree Tensor Network States (TTNS) — decompose weight matrices into low-rank components, enabling 2–10x compression while preserving accuracy. Demonstrated in CompactifAI for LLMs [1].
Quadratic Attention Complexity $O(N^2)$	Every token must interact with every other token in the sequence. Doubling the sensor history window quadruples compute and memory requirements. For an AD vehicle processing 8+ camera feeds plus LiDAR, or a robot with multi-modal sensors, this directly threatens the 100ms real-time safety deadline.	Quantum-inspired low-rank attention approximations or quantum kernel methods could reduce this to $O(N \log N)$ or better, making longer context windows feasible without sacrificing real-time performance.
RL Alignment Training Cost	GRPO-style reinforcement learning requires sampling multiple completions per training instance to estimate gradients. With a 7B–10B backbone, this multiplies GPU-hours significantly. Each policy update for an AD driving policy or a robot manipulation skill is expensive and sample-inefficient.	Quantum Reinforcement Learning (QRL) circuits or parameterized quantum circuit (PQC) policy heads may improve sample efficiency — reaching equivalent reward quality with fewer rollouts, reducing the RL alignment compute budget substantially [2].
Control Safety Guarantees	Current safety reward signals are heuristic indicators, not engineering constraints. An AD vehicle or robot arm operating under sensor noise or distribution shift has no mathematical guarantee it will converge to a safe state. This is a barrier to regulatory certification under ISO 26262 and IEC 62061.	Quantum or quantum-inspired Lyapunov stability analysis can provide formal proof of safe convergence. A Lyapunov function satisfying $V(x) > 0$ and $dV/dt \leq 0$ guarantees the system always moves toward a safe equilibrium, even under disturbances — replacing heuristic clipping with a verifiable certificate.

5.2 Required Outputs

All submissions must include:

- A functional quantum-enhanced or quantum-inspired component integrated into at least one stage of VLAM training, compression, or control
- A public code repository with a README enabling clean-environment replication of all reported results, including dependencies, random seeds, and hardware specs
- Quantitative results vs. the track-specific accepted baseline, reported as mean ± standard deviation across at least 3 independent runs
- A 4-8 page technical report covering: method description, theoretical motivation, experimental setup, results, and an ablation isolating the quantum/QI component from the rest of the pipeline

5.3 Constraints

Physical quantum hardware access is not required and is not provided by the organizers. Simulation-based demonstrations are fully acceptable and evaluated on equal terms with hardware-based submissions. Participants may use any quantum hardware or simulation environment provided all resource assumptions (qubit count, gate fidelity, simulation environment name and version, total shot budget) are explicitly stated. Any degradation in task accuracy or safety performance relative to the baseline must be fully reported and discussed in the technical report.

5.4 Accepted Baselines

Baselines are defined to ensure fair, domain-appropriate comparison. Participants may use the baselines below for reference. However, any alternative approaches or methods may be used along with a brief justification. A general overview of the possibilities is given below.

Table on following page.

Application Context 1: Autonomous Driving	Application Context 2: Robotics
Reference model: Alpamayo-R1 [3] or LLaVA-1.5-7B	**Reference model:** π_0 (Pi-Zero) [7] or OpenVLA-7B
Reference datasets: nuScenes [6], CARLA, Waymo Open Dataset	**Reference datasets:** Open X-Embodiment [5], RLBench, LIBERO
Safety standard: ISO 26262 (automotive functional safety)	**Safety standard:** IEC 62061 (machinery/robotics), ISO 10218
Deployment target: NVIDIA or Qualcomm SoC, ≤100ms inference	**Deployment target:** Embedded robot controller, ≤100ms inference

Application Context 1: Autonomous Driving

Challenge Sub-Track	Reference Model	Reference Dataset	Accepted Baseline Method	What Is Measured
Compression	LLaVA-1.5-7B	nuScenes or Waymo	INT8 quantization via bitsandbytes	Task accuracy on held-out split; total parameter count
Training Efficiency	Alpamayo-R1 [3]	CARLA simulation	AdamW optimiser + cosine LR decay	Training steps to convergence; total GPU-hours consumed
RL Alignment	Any 7B–10B VLAM	CARLA or Waymo AD scenarios	Standard GRPO rollout budget (no QI/QRL)	Policy reward at convergence; total rollouts consumed
Safety Guarantees	Any AD VLAM control policy	Synthetic perturbation suite	Heuristic reward clipping at path boundary	% of episodes terminating in a safe equilibrium state

Application Context 2: Robotics

Challenge Sub-Track	Reference Model	Reference Dataset	Accepted Baseline Method	What Is Measured
Compression	π_0 (Pi-Zero) [7]	Open X-Embodiment [5]	INT8 quantization via bitsandbytes	Task accuracy on held-out split; total parameter count
Training Efficiency	π_0 (Pi-Zero) [7]	RLBench suite	AdamW optimiser + cosine LR decay	Training steps to convergence; total GPU-hours consumed
RL Alignment	Any 3B–10B VLAM	RLBench or LIBERO tasks	Standard GRPO rollout budget (no QI/QRL)	Policy reward at convergence; total rollouts consumed
Safety Guarantees	Any robot VLAM control policy	Synthetic perturbation suite	Heuristic reward clipping at joint/velocity limit	% of episodes terminating in a safe equilibrium state

5.5 Performance Benchmarks

The following performance benchmarks define what a strong solution looks like for the VW challenge. All metrics must be reported with mean and standard deviation across at least 3 independent runs. Note: overall participant selection is governed by the challenge-wide assessment criteria published separately by Resonance / The Quantum Insider.

Benchmark	How It Is Measured	Guidance threshold
Efficiency Gain	FLOPs reduction (%) or wall-clock training time reduction compared to the accepted baseline. Reported as mean ± standard deviation across at least 3 independent runs.	≥10% improvement over baseline

Benchmark	How It Is Measured	Guidance threshold
Compression vs. Accuracy	Parameter reduction ratio paired with task accuracy change on the track-specific benchmark. If multiple compression levels are tested, submit a Pareto curve showing the tradeoff.	≤5% accuracy drop at ≥2x compression
Latency on Reference Profile	End-to-end inference time (ms) measured on a clearly stated hardware profile (e.g., CPU-only edge device or quantum simulator). Must stay within the real-time safety budget for the chosen track.	≤100ms on stated hardware profile
Reproducibility	Submitted code must run end-to-end from a clean environment with no manual intervention. All baselines, datasets, hyperparameters, random seeds, and hardware specs must be documented in the repository.	Full clean-environment replication required
Quantum Justification	A clear theoretical or empirical explanation of why the quantum or QI component provides the observed advantage. Must include an ablation study that isolates the quantum/QI component from the rest of the pipeline.	Ablation study is mandatory

Note on Accuracy-Efficiency Tradeoffs

Judges understand that efficiency gains often involve some accuracy cost. Submissions are not penalized for accuracy degradation below the stated minimum threshold, provided the tradeoff is clearly characterized.

6. Data and Resources

No proprietary data, compute, quantum hardware access, or cloud credits are provided by the organizers. Participants must independently source all resources. Participants focused on Autonomous Driving applications can refer to nuScenes [6], Waymo Open Dataset, and CARLA; and participants focused on Robotics applications can refer to Open X-Embodiment [5], RLBench, and LIBERO. All datasets must be used in compliance with their respective licences.

Freely available recommended tools: Qiskit, PennyLane, and Cirq for quantum circuit simulation; ITensor, quimb, and tntorch for tensor network compression; PyTorch, JAX, DeepSpeed, and FSDP for classical training. All submissions must include a Resource Declaration (GPU type and count, total GPU-hours, simulation environment and version, estimated energy consumption) for reproducibility verification. This declaration has no bearing on evaluation scores.

7. References

[1] Tomut, A. et al. (2024). CompactifAI: Extreme Compression of Large Language Models using Quantum-Inspired Tensor Networks. arXiv:2401.14109.

[2] Liu, J. et al. (2024). Towards Provably Efficient Quantum Algorithms for Large-Scale Machine-Learning Models. Nature Communications, 15(1), 434.

[3] NVIDIA & Wang, Y. et al. (2025). Alpamayo-R1: Bridging Reasoning and Action Prediction for Generalizable Autonomous Driving in the Long Tail. arXiv:2511.00088.

[4] NVIDIA Technical Blog (2025). Introducing NVFP4 for Efficient and Accurate Low-Precision Inference. NVIDIA Developer Blog.

[5] Open X-Embodiment Collaboration (2023). Open X-Embodiment: Robotic Learning Datasets and RT-X Models. arXiv:2310.08864.

[6] Caesar, H. et al. (2020). nuScenes: A Multimodal Dataset for Autonomous Driving. CVPR 2020, pp. 11618–11628.

[7] Black, K. et al. (2024). π_0: A Vision-Language-Action Flow Model for General Robot Control. arXiv:2410.24164.